

The $SU(2)$ bridge midpoint in closed form: the curvature term is exact

Result, 2026-09-27. Proposition 6 of the series/parallel note described the midpoint of the Brownian bridge on $SU(2)$, the building block of the non-abelian parallel insertion, only to leading order in the heat time, with the curvature factor $1/h$, $h = (d/4) \cot(d/4)$, derived by Laplace's method and labelled as asymptotics. For the fundamental character the midpoint expectation is computable exactly. For a bridge of duration t from e to a group element at geodesic distance $\theta \in (0, 2\pi)$,

$$E \chi_{1/2}(m) = e^{-t/32} \left[2 \cos \frac{\theta}{4} - \frac{t}{2\theta} \sin \frac{\theta}{4} \right] \text{ up to image terms of relative size } \left(2 + \frac{32\pi^2}{t} \right) e^{-4\pi(2\pi-\theta)/t}$$

(Theorem 1). Expanded to first order it is exactly the prediction of Proposition 6(b), $2 \cos \frac{\theta}{4} [1 - \frac{t}{32} (1 + \frac{2}{h(\theta)})]$, because $\frac{t}{32} \cdot \frac{2}{h} = \frac{t}{4\theta} \tan \frac{\theta}{4}$. So the curvature term N2, the side-flux-dependent softening of the mid-face weight, is confirmed with no remainder of order t^2 in the $w = 0$ sector: the only correction to the displayed formula is the exponentially small winding of the bridge around the group. This is the first exact non-abelian statement about the parallel insertion in three dimensions, and it settles the fundamental-representation part of the uniform Laplace remainder that Proposition 7 of the series/parallel note assumes, for the trace in the fundamental representation (the image terms are of the stated order, and not uniformly so as $\theta \rightarrow 2\pi$).

The ingredients are character orthogonality and Poisson summation; the computation is elementary and no novelty is claimed for it.

1. Setting

Group metric as in the series/parallel note §1: $C_2(j) = j(j+1)$, $SU(2)$ the three-sphere of radius 2, heat kernel $k_t = \sum_j (2j+1) e^{-tj(j+1)/2} \chi_j$ with $\chi_j(\theta) = \sin((2j+1)\theta/2) / \sin(\theta/2)$ at rotation angle θ (the geodesic distance from e). The midpoint m of the bridge of duration t from e to g has density $k_{t/2}(m) k_{t/2}(m^{-1}g) / k_t(g)$ with respect to Haar measure. Its geodesic midpoint $m_* = \exp(\frac{1}{2} \log g)$ has rotation angle $\theta/2$, so $\chi_{1/2}(m_*) = \sin(\theta/2) / \sin(\theta/4) = 2 \cos(\theta/4)$.

2. The exact formula

Theorem 1. Put $\Theta_{\pm}(\theta) = \sum_{w \in \mathbb{Z}} (\pm 1)^w e^{-(\theta - 4\pi w)^2 / (2t)}$ and $\Xi_{\pm}(\theta) = \sum_{w \in \mathbb{Z}} (\pm 1)^w (\theta - 4\pi w) e^{-(\theta - 4\pi w)^2 / (2t)}$. For $\theta \in (0, 2\pi)$,

$$E \chi_{1/2}(m) = e^{-t/32} \frac{2 \cos(\theta/4) \Xi_{-}(\theta) - \frac{t}{2} \sin(\theta/4) \Theta_{-}(\theta)}{\Xi_{+}(\theta)}. \quad (1)$$

Keeping only $w = 0$ gives the displayed formula of the summary.

Proof. Expand both heat kernels in characters. Character orthogonality in the form $\int \chi_a(m) \chi_b(m^{-1}g) dm = \delta_{ab} \chi_a(g) / d_a$, together with $\chi_{j_1} \chi_{1/2} = \chi_{j_1+1/2} + \chi_{j_1-1/2}$, gives

$$k_t(g) E \chi_{1/2}(m) = \sum_{j_1} \sum_{j_2=j_1 \pm 1/2} (2j_1+1) e^{-t[C_2(j_1)+C_2(j_2)]/4} \chi_{j_2}(g).$$

Write $n = 2j_1 + 1$. Then $C_2(j_1) + C_2(j_2) = \frac{1}{4}[n^2 + (n \pm 1)^2 - 2] = \frac{1}{2}(n \pm \frac{1}{2})^2 - \frac{3}{8}$, so the exponential is $e^{3t/32} e^{-t(n \pm 1/2)^2/8}$. Multiply by $\sin(\theta/2)$, reindex the $j_2 = j_1 - \frac{1}{2}$ sum by $n \mapsto n+1$, and write $s = n + \frac{1}{2}$ over $s \in \frac{1}{2} + \mathbb{N}$: the summand becomes $(s - \frac{1}{2}) \sin \frac{(s+\frac{1}{2})\theta}{2} + (s + \frac{1}{2}) \sin \frac{(s-\frac{1}{2})\theta}{2} = 2s \cos \frac{\theta}{4} \sin \frac{s\theta}{2} - \sin \frac{\theta}{4} \cos \frac{s\theta}{2}$, which is even in s . Hence

$$k_t(g) \sin \frac{\theta}{2} E \chi_{1/2}(m) = \frac{e^{3t/32}}{2} \left[2 \cos \frac{\theta}{4} A - \sin \frac{\theta}{4} B \right],$$

with $B = \sum_{s \in \frac{1}{2} + \mathbb{Z}} e^{-ts^2/8} \cos \frac{s\theta}{2}$ and $A = \sum_{s \in \frac{1}{2} + \mathbb{Z}} s e^{-ts^2/8} \sin \frac{s\theta}{2} = -2\partial_\theta B$. In the same way $k_t(g) \sin \frac{\theta}{2} = \frac{e^{t/8}}{2} A_0$ with $A_0 = \sum_{n \in \mathbb{Z}} n e^{-tn^2/8} \sin \frac{n\theta}{2}$. Poisson summation over the half-integers (which introduces the sign $(-1)^w$) and over the integers gives $B = \sqrt{8\pi/t} \Theta_-$, $A = (2/t) \sqrt{8\pi/t} \Xi_-$ and $A_0 = (2/t) \sqrt{8\pi/t} \Xi_+$. Divide. $\square$

The image terms. For $\theta \in (0, 2\pi)$ the pair $w = \pm 1$ in $\Xi_\pm$ has, relative to the $w = 0$ term $\theta e^{-\theta^2/(2t)}$, the size $e^{-8\pi^2/t} |2\theta \cosh(4\pi\theta/t) - 8\pi \sinh(4\pi\theta/t)|/\theta \leq (2 + 32\pi^2/t) e^{-4\pi(2\pi-\theta)/t}$, using $\sinh x \leq x \cosh x$. In Θ_- the same pair has relative size at most $2e^{-4\pi(2\pi-\theta)/t}$. The pairs $|w| \geq 2$ are smaller by a further factor $e^{-16\pi^2/t}$ up to polynomial factors. This is the large-field term N4, the winding of the bridge around the group, and it is negligible uniformly on $\theta \leq \theta_0 < 2\pi$.

3. Comparison with Proposition 6

Proposition 6(b) predicts that $m = m_* e^\xi$ with ξ Gaussian of mean zero and co-variance $\frac{t}{4} \text{diag}(1, \frac{1}{h}, \frac{1}{h})$ in the frame (axis, transverse). In the spin- $\frac{1}{2}$ representation $(\xi \cdot T)^2 = -\frac{1}{4} |\xi|^2$, so $E D^{1/2}(e^\xi) = 1 - \frac{1}{8} E |\xi|^2 + O(t^2) = 1 - \frac{t}{32} (1 + \frac{2}{h}) + O(t^2)$, and $E \chi_{1/2}(m) = 2 \cos \frac{\theta}{4} [1 - \frac{t}{32} (1 + \frac{2}{h})] + O(t^2)$. The $w = 0$ part of (1) equals $2 \cos \frac{\theta}{4} e^{-t/32} [1 - \frac{t}{4\theta} \tan \frac{\theta}{4}]$, and $\frac{t}{32} \cdot \frac{2}{h(\theta)} = \frac{t}{16} \cdot \frac{\tan(\theta/4)}{\theta/4} = \frac{t}{4\theta} \tan \frac{\theta}{4}$. The two agree at first order, curvature factor included. As $\theta \rightarrow 0$ both give $2(1 - \frac{3t}{32})$, the flat value $E |\xi|^2 = \frac{3t}{4}$.

Formula (1) says more than the asymptotics. In the $w = 0$ sector the fundamental character of the midpoint is the product of an isotropic factor $e^{-t/32}$ and the exactly linear curvature correction $1 - \frac{t}{4\theta} \tan \frac{\theta}{4}$; there are no higher-order terms in t . The correction grows as $\theta \rightarrow 2\pi$, where $\tan(\theta/4)$ diverges and the midpoint becomes bimodal, and there the image terms take over. This is the non-abelian counterpart of the parity factor $\rho_t(\phi)$ of the $U(1)$ cube (Proposition 3 of the series/parallel note).

4. Every spin

Theorem 2. For every spin J and $\theta \in (0, 2\pi)$, with $\sigma = 0$ for integer J and $\sigma = \frac{1}{2}$ for half-integer J ,

$$E \chi_J(m) = \frac{\Xi_{(\sigma)}(\theta)}{\Xi_+(\theta)} \sum_{k=-J}^J e^{-tk^2/8} \cos \frac{k\theta}{2} - \frac{t}{2} \frac{\Theta_{(\sigma)}(\theta)}{\Xi_+(\theta)} \sum_{k=-J}^J k e^{-tk^2/8} \sin \frac{k\theta}{2},$$

where $\Xi_{(0)} = \Xi_+$, $\Theta_{(0)} = \Theta_+$, $\Xi_{(1/2)} = \Xi_-$, $\Theta_{(1/2)} = \Theta_-$, and k runs in unit steps. In the $w = 0$ sector,

$$E \chi_J(m) = \sum_{k=-J}^J e^{-tk^2/8} \left[\cos \frac{k\theta}{2} - \frac{t}{2\theta} k \sin \frac{k\theta}{2} \right]. \quad (2)$$

For integer J the first ratio is exactly one, image terms included.

Proof. As for Theorem 1, now with $\chi_{j_1} \chi_J = \sum_{k=-J}^J \chi_{j_1+k}$. This identity holds for every j_1 if χ_j is defined by $\sin((2j+1)\theta/2)/\sin(\theta/2)$ also for $2j+1 \leq 0$: it is the product-to-sum formula for $\sin(n\theta/2) \sum_k e^{ik\theta}$. The formal terms with $2j_2+1 \leq 0$ cancel in pairs carrying the same weight, so the sum may be taken over all $n = 2j_1+1 \in \mathbb{Z}$, halved; the summand is even under $(n, k) \mapsto (-n, -k)$. With $n_2 = n + 2k$, $C_2(j_1) + C_2(j_2) = \frac{1}{2}(n+k)^2 + \frac{1}{2}(k^2-1)$, so the weight is $e^{t/8} e^{-tk^2/8} e^{-ts^2/8}$ with $s = n+k \in \mathbb{Z} + \sigma$. Expanding $(s-k) \sin \frac{(s+k)\theta}{2}$ and discarding the terms odd in s leaves $\cos \frac{k\theta}{2} \cdot s \sin \frac{s\theta}{2} - k \sin \frac{k\theta}{2} \cos \frac{s\theta}{2}$. Summing over s gives A_σ and B_σ ,

Poisson summation gives $\Xi_{(\sigma)}$ and $\Theta_{(\sigma)}$, and the factor $e^{t/8}$ cancels against the denominator. For $J = \frac{1}{2}$ this is Theorem 1. $\square$

Agreement with Proposition 6 for every spin. The leading term of (2) is $\sum_k \cos(k\theta/2) = \chi_J(m_*)$, the character at the geodesic midpoint (rotation angle $\theta/2$). The Gaussian prediction of Proposition 6(b) at first order is $\sum_k \cos \frac{k\theta}{2} [1 - \frac{t}{8}k^2 - \frac{t}{8}(J(J+1) - k^2)/h]$ with $1/h = 4 \tan(\theta/4)/\theta$. The two agree iff, with $\varphi = \theta/2$,

$$\sum_k k \sin(k\varphi) = \tan \frac{\varphi}{2} \sum_k (J(J+1) - k^2) \cos(k\varphi).$$

The left side is $-\chi'_J(\varphi)$ and the sum on the right is $J(J+1)\chi_J + \chi''_J$. The class-function Laplacian of $SU(2)$ in this metric gives $\chi''_J + \cot(\varphi/2)\chi'_J = -J(J+1)\chi_J$, which is the identity. (For $J = \frac{1}{2}$ and $J = 1$ it can be checked directly: $\sin(\varphi/2) = \tan(\varphi/2)\cos(\varphi/2)$ and $2\sin\varphi = \tan(\varphi/2)(2 + 2\cos\varphi)$.) So the curvature term of Proposition 6 is exact at first order in every representation, and (2) contains no terms of order t^2 beyond the Gaussian factors $e^{-tk^2/8}$.

What the trace does not give. By Proposition 6(a), $ED^J(m) = D^J(m_*)\Lambda^J$ with Λ^J real diagonal in the axis basis, and (2) fixes only $\sum_\mu e^{i\mu\theta/2}\Lambda_{\mu\mu}^J$, one equation for $2J+1$ entries. The isolated cube needs the entries, because its four cut faces have different axes.

5. The fundamental sector of the isolated cube is exact

Theorem 3. (a) In spin $\frac{1}{2}$ the midpoint's matrix expectation is a real multiple of a unitary: $ED^{1/2}(m) = \lambda(\theta, t) D^{1/2}(m_*)$ with

$$\lambda(\theta, t) = e^{-t/32} \frac{\Xi_-(\theta) - \frac{t}{4} \tan \frac{\theta}{4} \Theta_-(\theta)}{\Xi_+(\theta)} \stackrel{w=0}{=} e^{-t/32} \left[1 - \frac{t}{4\theta} \tan \frac{\theta}{4} \right], \quad \theta \in (0, 2\pi).$$

(b) For one refined cube (series/parallel note, §4), whose four cut side faces have heat times t_j and side angles θ_j , the spin- $\frac{1}{2}$ term of the character sum for Ψ is exactly

$$d_{1/2} e^{-3tm/8} \text{tr}_{1/2} \prod_j ED^{1/2}(m_j^{\pm 1}) = 2 e^{-3tm/8} \left(\prod_{j=1}^4 \lambda(\theta_j, t_j) \right) \chi_{1/2}(h_{\text{int}}),$$

where h_{int} is the holonomy of the geodesic interpolation around the mid-face. In this sector the interpolation term N1 is exact (it is h_{int}), the curvature term N2 is exact (it is $\prod \lambda$), and the commutator term N3 vanishes.

Proof. (a) By Proposition 6(a) of the series/parallel note, $ED^{1/2}(m) = D^{1/2}(m_*)\Lambda$ with $\Lambda = \text{diag}(\Lambda_+, \Lambda_-)$ real in the axis basis, where $D^{1/2}(m_*) = \text{diag}(e^{i\theta/4}, e^{-i\theta/4})$. Its trace $e^{i\theta/4}\Lambda_{++} + e^{-i\theta/4}\Lambda_{--}$ equals $E\chi_{1/2}(m)$, which is real because the midpoint density and $\chi_{1/2}$ are real. The imaginary part gives $\sin(\theta/4)(\Lambda_+ - \Lambda_-) = 0$, so $\Lambda_+ = \Lambda_- = \lambda$, and $\lambda = E\chi_{1/2}(m)/(2\cos(\theta/4))$ by Theorem 1. (b) The bridge for a cut face runs from Q_j^{-1} to P_j ; left translation by Q_j^{-1} maps it to a bridge from e , so $ED^{1/2}(m_j) = \lambda_j D^{1/2}(m_{*j})$ with m_{*j} the geodesic midpoint, and $ED^{1/2}(m_j^{-1}) = \lambda_j D^{1/2}(m_{*j})^{-1}$ because λ_j is real. The four midpoints are independent, since each is fixed by its own cut face and an isolated cube has no shared mid-edges, so the trace of the product of expectations is $\prod_j \lambda_j$ times the trace of the product of the unitaries, which is $\chi_{1/2}(h_{\text{int}})$. The prefactor is $d_{1/2} e^{-tmC_2(1/2)/2}$ with $C_2(\frac{1}{2}) = \frac{3}{4}$. $\square$

For $J \geq 1$ the diagonal entries differ: the Gaussian prediction is $\Lambda_{\mu\mu}^J \simeq 1 - \frac{t}{8}[\mu^2 + (J(J+1) - \mu^2)/h]$, which depends on μ unless $h = 1$. The Weyl reflection combined with time reversal gives $\Lambda_{\mu\mu}^J = \Lambda_{-\mu, -\mu}^J$ for every J : with w in the normaliser of the torus, $\text{Ad}(w)X = -X$, the map $g \mapsto e^X w g w^{-1}$ carries the bridge from e to e^X to the bridge from e^X to e , whose midpoint law is the same by reversibility, and conjugation by $D(w)$ exchanges μ and $-\mu$ on a diagonal

matrix (reviewer's argument; the reality of the trace alone gives only one relation). The trace (2) then fixes $\lfloor J \rfloor + 1$ unknowns by one equation, so for $J = 1$ one relation remains between Λ_{11}^1 and Λ_{00}^1 . The matrix-element version of the orthogonality computation of Theorem 1, with the Clebsch–Gordan coefficients of $j_1 \otimes J$, is the tool for them.

6. Consequence for STATE

The curvature term of the $SU(2)$ parallel insertion is now exact for every character (Theorems 1 and 2), with only exponentially small winding terms left, and the spin- $\frac{1}{2}$ sector of the isolated cube is exact (Theorem 3). One step remains before Proposition 7 becomes a theorem for the isolated $SU(2)$ cube: the diagonal entries $\Lambda_{\mu\mu}^J$, $J \geq 1$, of the midpoint's matrix expectation, since the cube's character sum $\sum_J d_J e^{-t_m C_2(J)/2} \text{tr}_J \prod_j E D^J(m_j^{\pm 1})$ multiplies matrices with different axes. They follow from the same orthogonality method applied to matrix elements, or from the characters of the midpoint translated along the centralizer of the side holonomy.